MIL-P-51227B(MU)
27 June 1972
SUPERSEDING
MIL-P-51227A(MU)
5 September 1969

MILITARY SPECIFICATION

PUMP, CENTRIFUGAL, DECONTAMINATING APPARATUS

1. SCOPE

1.1 This specification covers a pump for a decontaminating apparatus.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-D-12468 - Decontaminating Agent, STB.
MIL-A-51027 - Antisetting Compound, Decontaminating Slurry, M2.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL5-45-2835 - Pump Assembly.

(Copies of specifications standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4230

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN INSTITUTE OF CHEMICAL ENGINEERS

Pump Manual

(Copies may be obtained from the American Institute of Chemical Engineers, 25 West 45th Street, New York 36, New York.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-45-2835 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings. Material not definitely specified shall be of a quality regularly used by manufacturers of pumping equipment.

3.1.2 Components. All components of the pump shall conform to the specifications and drawings listed on Drawing DL5-45-2835 and subsidiary drawings.

3.2 Assembly. The pump shall be assembled as shown on Drawing D5-45-2835.

3.3 Functioning. The pump shall meet the following requirements when running at 4400 ± 100 revolutions per minute (rpm) for a minimum period of 15 minutes when tested as specified in 4.4.4.1.

(a) Discharge 100 gallons per minute (gpm) minimum of water at a discharge head of 205 ± 1 feet with a minimum static lift of 15 feet.
(b) Operate smoothly and not leak.
(c) Prime in a maximum of 4 minutes by using no more than 9 gallons of water.

3.4 Performance. The pump running at 4400 + 100 rpm shall be capable of delivering a minimum of 50 gpm of STB slurry, made in accordance with 4.4.4.2, at a discharge head of 290 ± 1 foot and a minimum dynamic lift of 10 feet. The pump shall also deliver a minimum of 100 gpm of water at a discharge head of 205 + 1 foot with a minimum static lift of 15 feet when running at 4400 + 100 rpm. The total head at zero flow shall be a

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minimum of 300 feet. The required horsepower (hp) to drive the pump shall not exceed 20 hp when running at 4400 ± 100 rpm and pumping a minimum of 50 gpm of STB slurry or an equivalent mass flow of water at a minimum of 140 pounds per square inch (psi) at the dishcharge gage.

3.5 Endurance. The pump shall be operated for a minimum of 250 hours with only minor adjustments. The pump shall operate smoothly without overloading or leakage when tested as specified in 4.4.4.3.

3.6 Performance curve. The discharge head versus capacity curve of the pump when tested as specified in 4.4.4.4 shall slope continuously from high head downward toward high capacity. The curve may approach constant head as the capacity increases but the head may never increase. The head capacity coordinates indicated in 3.3 shall be on or below the pump performance curve.

3.7 Cavitation. Cavitation shall not occur when tested as specified in 4.4.4.5.

3.8 Weight. The assembled pump weight shall not exceed 100 pounds when tested as specified in 4.4.4.6.

3.9 Preproduction. Prior to the start of regular production, a preproduction sample of pumps shall be produced for examination and tests in accordance with 4.3.

3.10 Workmanship. The pump shall be free from contamination (foreign matter, oil or grease) and damage such as cracks, burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and

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section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of two pumps shall be produced using the same methods, materials, and equipment that will be used in regular production.

4.3.2 Inspection procedure. The pumps shall be tested in accordance with 4.4.4.2, 4.4.4.4, and 4.4.4.6. One pump shall also be tested in accordance with 4.4.4.3 and 4.4.4.5. If that pump fails these tests, cause of failure shall be determined by the contractor and the second pump test in accordance with 4.4.4.3 and 4.4.4.5.

4.3.3 Acceptance/rejection criteria. At least one of the preproduction pumps shall meet all the tests of this specification to be acceptable. The supplier shall obtain written approval from the contracting officer prior to starting regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the pumps of one interfix number, produced by one manufacturer, at one plant, for the same materials, and under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 Examination and nondestructive tests. Sample pumps shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 Critical characteristics. Each pump in the lot shall be examined for the critical characteristic listed in the classification of defects.

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4.4.3.3 Classification of defects.

(a) Pump assembly (Dwg D5-45-2835).

Categories	Defects	Acceptance standards
Critical:		
1	Functioning (less prime time test)	4.4.4.1
Major:	AQL 1.0 percent defective	
101	Grease fitting missing	
102	Diameter of mounting holes incorrect (Dwg E5-45-2756)	
103	Location of mounting holes incorrect (Dwg E5-45-2756)	
104	Thread (2-11-1/2NPT) incorrect (Dwg E5-45-2756)	
105	Thread (1/2-14NPT) incorrect (Dwg E5-45-2756)	
106	Thread (3/8-18NPT) incorrect (Dwg D5-45-2776)	
107	Shaft diameter (0.875) incorrect (Dwg C5-45-2751)	
108	Keyway depth (0.245), width (0.2487) and radius (0.887) incorrect (Dwg. C5-45-2751)	
109	Marking missing or incorrect (Dwg E5-45-2756)	
110	Breather obstructed	
111	Workmanship (3.10)	
112	Prime time test	4.4.4.1

4.4.4 Tests.

4.4.4.1 Functioning. The pump shall be operated at the required speed for the specified time pumping water. Test shall be conducted in accordance with procedures outlined in section 8 of the American Institute of Chemical Engineers Centrifugal Pump Manual. The pump shall be observed for smoothness of operation and leakage. Discharge capacity, discharge head, static lift and time required to prime the pump shall be recorded.

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4.4.4.2 Performance. The performance of the pump shall be as specified when connected to a suitable drive and supply tank. The horsepower shall be determined when pumping slurry and tested in accordance with section 8 of the American Institute of Chemical Engineers Centrifugal Pump Manual. The slurry, when required for the above tests, shall have the following proportions:

(a) Two hundred and twenty-five gallons of water.
(b) Twelve and one half pounds of decontaminating slurry antiset M2 conforming to MIL-A-51027 (antiset shall be added to the water before STB).
(c) Thirteen hundred pounds of decontaminating agent STB conforming to MIL-D-12468.

4.4.4.3 Endurance. After completion of the test in 4.4.4.2 the pump shall be operated pumping slurry from and to a tank. Fresh slurry mixture shall be prepared whenever required, but at least every 16 hours. The pump shall be flushed and cleaned whenever shutdown for more than 1 hour. The pump shall be run at 4400 ± 100 rpm and discharge the slurry through an orifice 0.562 + 0.010 inch in diameter. Necessary precautions shall be taken to keep the temperature of the mixture being pumped between 85° F and 135° F while testing. The slurry shall have the following proportions for the above test:

(a) Two hundred and twenty-five gallons of water.
(b) Twenty-four ounces of Dow Antifoam B .
(c) Twelve and one half pounds of decontaminating slurry antiset M2 conforming to MIL-A-51027 (the antiset shall be added to the water before the STB).
(d) Six hundred and fifty pounds of decontaminating agent STB conforming to MIL-D-12468.

4.4.4.4 Performance curve. Performance of the pump shall be determined using plain water. Test shall be conducted in accordance with procedures outlined in section 8 of the American Institute of Chemical Engineers Centifugal Pump Manual. A minimum of six points including no flow and maximum flow conditions shall be obtained for the discharge head-capacity curve.

4.4.4.5 Cavitation. The impeller shall be visually examined for cavitation after testing as specified in 4.4.4.3. If necessary for examination, the impeller shall be cut perpendicular to the axis of rotation. If evidence of cavitation is present, the design is unsatisfactory and shall be rejected. If the impeller does not show evidence of cavitation and the pump meets the requirements of 3.7, the pump may be rebuilt to its new condition.

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4.4.4.6 Weight. The weight of the pump shall be as specified when weighed on a suitable scale accurate to within 1 pound.

5. PREPARATION FOR DELIVERY

5.1 Interplant shipment.

5.1.1 Packaging. The pump shall be cleaned, preserved and packaged in a manner that will afford protection and insure delivery of the item to a first receiving activity in an undamaged condition.

5.1.2 Packing. The pump, packaged in accordance with 5.1.1, shall be packed in a manner that will insure delivery of the item to a first receiving activity in an undamaged condition. Container types, sizes and load limits shall be in compliance with the rules and regulations of the applicable modes of transportation.

5.2 Repair parts shipment. When the pump is procured as a spare part, preservation, packaging and packing shall be in accordance with the respective packaging data sheet. (See 6.2).

6. NOTES

6.1 Intended use. This pump is intended for use with M9 or M12A1 decontaminating apparatus. The pump may also be used in other applications to pump various fluids.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation, packaging, and packing required.
(c) Preproduction.

(1) Time allowed for suppliers to submit samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instruction when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with regular production.

Custodian:

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Preparing activity:

Army - MU(EA)

Project No. 4230-A047

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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